MIL-R-28810(EC)
28 March 1975
SUPERSEDING (See 6.5)

MILITARY SPECIFICATION

RECEIVING SET, ELECTRONIC
COUNTERMEASURES, AN/WLR-11( )

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification establishes the performance, design, test and production requirements for Electronic Countermeasures Instantaneous Frequency Measuring (IFM) Radio Receiver AN/WLR-11( ) covering the frequency range of 7 to 18 gigahertz (GHz). The AN/WLR-11( ) receiver shall consist of the following units:

(a) Antenna AS-2918/S
(b) Amplifier, Radio Frequency AM-6799( )/WLR-11( )
(c) Radio Frequency Unit R-1814( )/WLR-11( ) consisting of three sub units:

(1) IFM receiver 7 - 11 GHz
(2) IFM receiver 11 - 18 GHz
(3) Power supply and switching assembly

(d) Two Control and Display Units C-9193( )/WLR-11 which are identical.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-E-16400	- Electronic, Interior Communication and Navigation Equipment, Naval Ship and Shore: General Specification for
MIL-P-17555	- Preparation for Delivery of Electronic Equipment, Miscellaneous Electrical Equipment (Except Rotating Electrical Equipment) and Associated Repair Parts

STANDARDS

MILITARY

MIL-STD-461	- Military Standard Electromagnetic Interference Characteristics Requirements for Equipment

FSC 5895

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MIL-STD-471	-	Maintainability Demonstration
MIL-STD-749	-	Preparation and Submission of Data for Approval of Non-Standard Electronic Parts
MIL-STD-781	-	Reliability Tests: Exponential Distribution

DRAWINGS

Receiving Set, Countermeasures AN/WLR-11A	-	Microfilm Roll No. 28606 NAVSEC Assignment Ltr Ser 1324-6181E.02 dtd 14 March 1975

PUBLICATIONS

NAVELEX 0967-LP-544-9010	-	Technical Manual for Countermeasures Receiving Set AN/WLR-11A
NAVELEX 0967-LP-544-9030	-	Technical Manual for Radio Frequency Amplifier AM-6799/WLR-11A and Antenna AS-2918/S

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. The AN/WLR-11( ) shall be identical in function to the AN/WLR-11A previously produced, as described in NAVELEX 0967-LP-544-9010 and NAVELEX 0967-LP-544-9030. The AN/WLR-11( ) shall be in accordance with the AN/WLR-11A Engineering Drawings on Microfilm Reel No. 28606 NAVSEC Assignment letter serial 1324-6181E.02. The AN/WLR-11( ) shall be comprised of the following units in accordance with the drawings specified in Table I.

Table I. AN/WLR-11A engineering drawings.

Unit	Reference drawing
Control Indicator C-9193( )/WLR-11( ) Radio Frequency Unit R-1814( )/WLR-11( ) Mast Amplifier, AM-6799( )/WLR-11( ) and Antenna AS-2918/S	Microfilm Reel No. 28606, NAVSEC Assignment letter Ser 1324-6181E.02

3.1.1 The equipment shall be in accordance with MIL-E-16400, except as hereinafter specified.

3.1.2 First article sample. Prior to beginning production, a sample shall be tested as specified in 4.3 (see 6.4).

3.1.3 Size and weight. The size and weight shall be as specified in NAVELEX 0967-LP-544-9010 and NAVELEX 0967-LP-544-9030.

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3.1.4 Modular construction requirements. The modular requirements of MIL-E-16400 are not applicable.

3.1.5 Environmental conditions.

3.1.5.1 Temperature. The units shall conform to the following temperature requirements of MIL-E-16400.

(a)	AS 2918/S	Class 2
(b)	AM/6799( )/WLR-11( )	Class 2
(c)	R-1814( )/WLR-11( )	Class 4
(d)	C-9193( )/WLR-11( )	Class 4

3.1.5.2 Vibration. Vibration performance shall conform to the requirements of MIL-E-16400. Maximum frequency of vibration shall be 33 Hertz (Hz).

3.1.5.3 Shock. Shock performance shall conform to the requirements of MIL-E-16400 for lightweight equipment, Class I, Type A.

3.1.5.4 Humidity. Humidity performance shall conform to the requirements of MIL-E-16400.

3.1.5.5 Inclination. The inclination requirements of MIL-E-16400 are not applicable.

3.1.5.6 Salt spray. The equipment shall conform to the salt spray testing requirements of MIL-E-16400.

3.1.5.7 Heat. Heat performance shall conform to the requirements of MIL-E-16400.

3.1.6 Enclosures.

3.1.6.1 The AS-2918/S and AM-6799( )/WLR-11( ) shall be fully enclosed in accordance with MIL-E-16400 and shall have a provision for dry-air pressurization.

3.1.6.2 The R-1814( )/WLR-11( ) and the C-9193( )/WLR-11( ) shall have drip-proof enclosures in accordance with MIL-E-16400.

3.1.7 Blown fuse indicators. Blown fuse indicators are not required.

3.1.8 Tropicalization. Tropicalization of the set is not required.

3.1.9 Time meter. An elapsed time meter shall be required.

3.1.10 Control knobs. Tactile identification of control knobs is not required.

3.1.11 Fuses and on-off switch(es). Fuses shall be of the miniature glass tube type and shall be mounted in insulated extractor type fuseholders. Each

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branch primary feeder circuit to power supplies of the major units of the equipment shall be separately fused. Both sides of the primary power circuit shall be opened by the ON-OFF switch.

3.1.12 Reliability. The AN/WLR-11( ) shall have a Mean-Time-Between-Failures (MTBF) of 1000 hours at a lower 90 percent confidence limit.

3.1.13 Maintainability. The AN/WLR-11( ) shall have a Median Equipment Repair Time (EKT) of 0.50 hours or less. The maximum time to repair within which 95 percent of all repairs are required to be accomplished, shall be 1.6 hours or less. These values apply to the correction of relevant failures only and shall be limited to repair at the organizational level.

3.1.14 Electromagnetic interference (EMI-EMC) requirements. The AN/WLR-11( ) shall be self-compatible and shall conform to the electromagnetic interference characteristics requirements of MIL-STD-461 for Class 1A, C-E equipment.

3.1.15 Interchangeability. The equipment shall conform to the interchangeability requirements of MIL-E-16400.

3.1.16 Safety requirement. The equipment shall conform to the safety design requirements of MIL-E-16400 except that the maximum shock hazard leakage current shall be 0.5 milliampere (mA).

3.1.17 Material. All material shall conform to the requirements of MIL-E-16400.

3.1.18 Parts. All parts to the extent possible shall be military standard in accordance with MIL-E-16400. Where nonstandard parts are required, the contractor shall comply with the requirements of MIL-STD-749 prior to the procurement of any part to be used in a deliverable equipment.

3.1.19 Equipment cooling. The equipment shall conform to the requirements of MIL-E-16400.

3.1.20 Power requirement. The equipment power factor shall be greater than 0.85. The individual harmonic distortion (IHD) shall be less than 3 percent up to the 23rd powerline harmonic; the 24th up to 20 kilohertz (kHz) falling off at a rate greater than $1700/n^2$.

3.2 Composition. The AN/WLR-11( ) shall be composed of the units and assemblies specified herein.

3.2.1 Antenna AS-2918/S. The AS-2918/S antenna shall provide the input for the AN/WLR-11( ). The antenna shall be a bi-conical type covering the range of 7 through 18 GHz. It shall be omni-directional in azimuth. It shall have 45 degree polarization, allowing reception of both horizontally and vertically polarized signals with a 3 decibels (dB) allowable loss.

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3.2.1.1 Antenna electrical characteristics. The antenna shall have the following electrical characteristics:

(a)	Frequency coverage	7 - 18 GHz
(b)	VSWR (50 ohm reference)	3:1 max
(c)	Gain (matched polarization)	0 dBi min
(d)	Azimuth pattern	Omni with 3 dB max ripple
(e)	Elevation pattern	3 dB beamwidth of 25 degrees min. with the maximum on the horizon.

3.2.2 Amplifier, radio frequency AM-6799( )/WLR-11( ). The AM-6799( )/WLR-11( ) amplifier shall provide amplification over the frequency range of 7 through 18 GHz. Its input shall be from antenna AS-2918/S and its outputs shall be to the receiver radio frequency (RF) Unit R-1814( )/WLR-11( ). The amplifier shall diplex the input into two frequency ranges, 7 to 11 GHz and 11 to 18 GHz (Low and High Bands). The diplexer shall have bandpass filtering characteristics for each band. The signal path for each band shall provide strong signal protection limiting and amplification. A capability to bypass each amplifier, by external command, shall be provided. This bypass shall be failsafe; for example, when power is lost the amplifiers shall be bypassed. The protection limiters shall remain in the signal path when the amplifiers are bypassed. An additional test input shall be provided which shall couple into the diplexer input.

3.2.2.1 Amplifier electrical characteristics. The amplifier shall have the following electrical characteristics.

		Low band (7 - 11 GHz)	High band (11 - 18 GHz)
(a)	Small signal gain	9 dB min.	13 dB min.
(b)	Noise figure	11.5 dB max.	14.5 dB max
(c)	Input and output VSWR (50 ohm ref.)	3:1 max.	3:1 max
(d)	Small signal gain variation vs. frequency +25 ±3°C	3 dB max.	5 dB max.
	-28 to 65°C	5 dB max.	8 dB max.
(e)	Saturated output power window	±-19 dBm	±-17 dBm
(f)	Saturated output power variation vs. frequency	±2 dB max.	±2 dB max.
(g)	Variation of transfer characteristic vs. temperature	±1.0 dB max.	2.0 dB max.

3.2.2.2 The requirements of 3.2.2.1 apply except as modified in the cross-over frequency area by the following diplexer requirements.

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	Low-band High-band	
(a)	Crossover frequency (F_{co})	11 GHz nominal
(b)	F_{co} tolerance 25° ±3°C	±55 megahertz (MHz) max
	-28° to +22°C	±100 MHz max
	+28 to +65°C	±100 MHz max
(c)	In-band insertion loss to within ±550 MHz of F_{co}	1 dB max
(d)	Insertion loss at F_{co}	5 dB max

3.2.3 RF unit, R-1814( )/WLR-11( ). The RF unit shall consist of two instantaneous frequency measuring (IFM) receiver modules, a power supply and switching module and associated cabling. These modules are to be mounted on slides enabling them to be withdrawn from the main frame with the power and control lines connected. Each RF module shall be a complete IFM receiver with its own amplifiers, YIGs, and frequency discriminator. They shall obtain their operating voltages from the power supply and switching module and signals from the AM-6799( )/WLR-11( ) preamplifier. Each RF module shall be controlled by direct current (DC) signals generated in a control and display unit, C-9193( )/WLR-11( ) and routed through the power supply and switching module. Video outputs from the RF modules shall be supplied to the control and display units via the power supply and switching modules as commanded by the Control and Display Unit. The two receiver modules and the power supply and switching module shall be mounted in a drip-proof enclosure which shall provide cabling to the rear panel, RFI line filters and overall RF interference integrity.

3.2.3.1 7 - 11 GHz receiver module. The 7 - 11 GHz receiver (low band) shall provide to the C-9193( )/WLR-11( ) the functions necessary to generate rectangular coordinate (X,Y) video signals as a function of input frequency. This shall be done simultaneously in two frequency bands, 7 - 11 GHz and 8.5-9.6 GHz (expand band). Broadband frequency discriminators shall be used as the basic frequency sensitive device. A undirectional video (Z), a function of signal strength, shall be generated for each signal. In addition, a 25 MHz wide, narrow band, YIG tuned radio frequency (TRF) channel shall operate in parallel with the broad band channel; it shall provide increased sensitivity on the display and in the video within the narrow frequency band to which it is tuned. The receiver shall be capable of operating in the narrow and broad band simultaneously (IFM/TRF), in narrow band only (TRF) or in the IFM only modes. A second RF input port for the application of test or marker signals shall be provided. Isolation shall be approximately 20 dB from the primary input. RF gain control shall be provided by a PIN diode attenuator which is controlled from the C-9193( )/WLR-11( ) Control and Display Unit. Pulse-by-pulse RF blanking capability shall be available. A notch filter (reject YIG) capable of being tuned over the complete band shall be provided within the primary RF signal path. A marker shall indicate the center frequency of this reject YIG on the face of the indicator oscilloscope when the reject YIG is turned ON. Band pass filtering capable of rejecting incoming out-of-band signals and internally generated spurious signals shall be provided. A TRF marking capability shall be provided at a RF which will generate, when commanded, a signal corresponding to the frequency to which the TRF portion of the Receiver is tuned. A 400 MHz signal shall be provided to a type BNC connector

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for external use to provide a WLR-1 band nine marking capability. A DC output signal shall be provided to the C-9193( )/WLR-11( ) to indicate that an in-band CW signal is being received.

3.2.3.2 11 - 18 GHz receiver module. The 11 - 18 GHz receiver (high band) shall provide to the C-9193( )/WLR-11( ) the functions necessary to generate rectangular coordinate (X,Y) video signals as a function of input frequency. A broadband frequency discriminator shall be used as the basic frequency sensitive device. A undirectional video (Z), a function of signal strength, shall be generated for each signal. In addition, a 30 MHz wide, narrow-band, YIG tuned, TRF channel shall operate in parallel with the broad band channel. It shall provide increased sensitivity on the display and in the video within the narrow frequency band to which it is tuned. The receiver shall be capable of operating in the narrow and broadband simultaneously (IFM/TRF) in narrow-band only (TRF) or in the IFM only modes. A second RF input port for the application of test or marker signals shall be provided. Isolation shall be approximately 20 dB from the primary input. RF gain control shall be provided by a PIN diode attenuator which is controlled from the C-9193( )/WLR-11( ) Control Display. Pulse-by-pulse RF blanking capability shall be available. A notch filter (reject YIG) capable of being tuned over the complete band shall be provided within the primary RF signal path. Bandpass filtering capable of rejecting incoming out-of-band signals and internally generated spurious signals shall be provided. A marker shall indicate the center frequency of the reject YIG on the face of the indicator oscilloscope when the reject YIG is turned ON. A TRF marking capability shall be provided at RF which will generate, when commanded, signals corresponding to the frequency to which the TRF portion of the receiver is tuned. A WLR-1 band ten marking capability shall be provided. A DC output signal shall be provided to the C-9193( )/WLR-11( ) which indicates that an in-band CW signal is being received.

3.2.3.3 Power supply and switching module. The power supply and switching module shall provide all DC voltages for itself and the two RF modules as well as serve as an interface between the control and display units and the RF modules. Front panel controls shall consist of a main alternating current (AC) power ON-OFF switch, a Remote-Local AC power control switch, and a switch controlling the preamplifier bypass for each RF band. Front panel indicators shall show that AC power is ON and whether by local or remote control. Line fuses shall be located on the panel. An elapsed time indicator showing total operating time shall be internally located. This module shall accept video signals from and provide control signals to the RF modules. It shall also decode band-selection signals and determine logic command priority for the two control and display units normally used in each system. It shall contain wideband video current-source line drivers for transmitting video to the control and display units.

3.2.3.4 Control and display unit, C-9193( )/WLR-11( ). The AN/WLR-11( ) receiving system shall have two identical control and display units. Each shall be capable of exercising control over all functions of the RF unit as well as processing and displaying video information supplied from the RF unit.

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3.2.3.4.1 Front panel. The front panel shall be integrally illuminated with variable intensity control. The display shall have a 5 inch (127.0 millimeters (mm)) cathode ray tube (CRT). It shall have electrostatic deflection and focus and type P7 phosphor. Three overlaying frequency calibrated graticules one for each position of the BAND switch, shall be provided. They shall be of clear plastic material with engraved frequency marking illuminated by variable intensity edge lighting; only one graticule shall be lit for each position of the BAND switch. Front panel controls shall be as specified in Table II.

Table II. Front panel controls.

Panel marking	Description or use
POWER	Illuminated push-button
BAND	3 positions 7-11, 8.5 - 9.6, 11 - 18
DEFL GAIN	2 position; NORM, HIGH
STRETCH	2 position, OFF, ON
PREAMP	2 position, OFF, ON
RF GAIN	Controls RF attenuation, with minimum attenuation at extreme clockwise position.
SCALE ILLUM	Adjust level of graticule illumination
FOCUS	Focus CRT beam
INTEN MOD	Adjust Z axis gain
AUDIO GAIN	Adjust headset audio level
PANEL DIM	Adjust intensity of indicator lights (except CW alarm) and the illuminated panel
LAMP TEST	Momentary push-button, allow test of indicator lights
AGU MODE	3 position; OFF, NORM, ALM
AGU WIDTH	Adjust position of CW cursor with respect to the CCW cursor
AGU POSITION	Adjust position of both cursors
RF MODE	2 position; IFM/TRF, TRF
MARK	3 position; WLR-1, OFF, TRF
SWP RATE	Adjust rate of bandpass YIG sweep, detent at extreme CCW sets manual YIG tune
BAND REJECT YIG TUNE	Adjust reject YIG tuning
BANDPASS YIG TUNE	Adjust Band-pass YIG tuning
CW ALARM	Indicates reception of a CW signal, flashes when activated
PRIORITY	On Control and Display Unit #2 indicates when Unit #1 has control of band selected, inactive on Unit #1
BAND REJECT	Lighted when reject YIG is tuned in to band
REJECT MARK	When turned ON-frequency of reject YIG is shown by cursor on indicator

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3.3 Electrical characteristics of the AN/WLR-11( ).

3.3.1 Frequency range. The frequency coverage of the AN/WLR-11( ) shall be 7 to 18 GHz. The frequency range shall be divided into two bands, 7 - 11 GHz and 11 - 18 GHz. A third band shall be an expansion of the 8.5 - 9.6 GHz range, within the 7 - 11 GHz band.

3.3.2 Modes of reception. There shall be three modes of reception of RF signals, (1) Wide-open channel (IFM), (2) tuned radio frequency (TRF), and (3) a combination of the above two in parallel (IFM/TRF). These modes shall be applicable to the particular band selected by the operator, with the mode of reception also selectable.

3.3.2.1 Wide open instantaneous frequency measuring (IFM) channel. Wide open (IFM) reception shall be by means of stripline discriminators and processing of resultant signals to present a frequency versus angular deflection indication on a CRT of all signals received. Coverage shall be of the entire applicable frequency band continuously so that all signals received in the pass-band are displayed simultaneously with an instantaneous indication of their frequency.

3.3.2.2 Tuned radio frequency (TRF) channel. A TRF channel shall be available in each band, with a 3 dB bandwidth of nominally 25 MHz in the 7 - 11 GHz and 8.5 - 9.6 bands and nominally 30 MHz in the 11 - 18 GHz band. The TRF channel shall be capable of being manually tuned, or being electronically tuned over the applicable frequency range of the band at a rate that may be varied by the operator from 0.1 sweep per second to approximately 50 sweeps per second.

3.3.3 Low-band (7-11 GHz) characteristics.

3.3.3.1 Sensitivity. All sensitivity measurements shall be made from the RF input of the receiver, not to include AM-6799( )/WLR-11( ) (see 6.3).

3.3.3.1.1 Sensitivity of the 7 - 11 GHz band wide open channel shall be not less than minus 60 dBm for normal (3/16 inch) (4.76 mm) deflection. High deflection gain sensitivity with minimum detectable signal shall be not less than minus 66 (dBm).

3.3.3.1.2 Sensitivity of the 7 - 11 GHz band narrow-band channel shall be not less than minus 69 dBm for normal (3/16 inch) (4.76 mm) deflection. High deflection gain sensitivity with minimum detectable signal shall be not less than minus 75 dBm.

3.3.3.2 Frequency resolution. Frequency resolution with normal deflection shall be not more than 20 MHz.

3.3.3.2.1 Frequency resolution with high deflection gain shall be not more than 40 MHz.

3.3.3.3 Frequency accuracy. Frequency accuracy with normal deflection gain shall be not more than 50 MHz.

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3.3.3.3.1 Frequency accuracy with high deflection gain shall be not more than 100 MHz.

3.3.4 Expanded band (8.5 - 9.6 MHz) characteristics.

3.3.4.1 Sensitivity. All sensitivity measurements shall be made from the RF input of the receiver, not to include the AM-6799( )/WLR-11( ).

3.3.4.1.1 Sensitivity of the 8.5 - 9.6 GHz band wide open channel shall be not less than minus 60 dBm for normal (3/16 inch) (4.76 mm) deflection. High deflection gain sensitivity with minimum detectable signal shall be not less than minus 60 dBm.

3.3.4.1.2 Sensitivity of the 8.5 - 9.6 GHz narrow-band channel shall be not less than minus 69 dBm for normal (3/16 inch) (4.76 mm) deflection. High deflection gain sensitivity with minimum detectable signal shall be not less than minus 78 dBm.

3.3.4.2 Frequency resolution. Frequency resolution with normal deflection shall be not more than 10 MHz.

3.3.4.2.1 Frequency resolution with high defelction gain shall be not more than 20 MHz.

3.3.4.3 Frequency accuracy. Frequency accuracy with normal deflection gain shall be not more than 15 MHz.

3.3.4.3.1 Frequency accuracy with high deflection gain shall be not more than 30 MHz.

3.3.5 High band (11 - 18 GHz) characteristics.

3.3.5.1 Sensitivity. All sensitivity measurements shall be made from the RF input of the receiver, not to include the AM-6799( )/WLR-11( ).

3.3.5.1.1 Sensitivity. Sensitivity of the 11 - 18 GHz band wide open channel shall be not less than minus 57 dBm for normal (3/16 inch) (4.76 mm) deflection. High deflection gain sensitivity with minimum detectable signal shall be not less than minus 63 dBm.

3.3.5.1.2 Sensitivity of the 11 - 18 GHz band narrow-band channel shall be not less than minus 66 dBm, for normal (3/16 inch) (4.76 mm) deflection. High deflection gain sensitivity with minimum detectable signal shall be not less than minus 72 dBm.

3.3.5.1.3 Sensitivity may degrade up to 3 dB above 16 GHz and up to 5 dB within 100 MHz of 18 GHz.

3.3.5.2 Frequency resolution. Frequency resolution with normal deflection shall be not more than 100 MHz.

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3.3.5.2.1 Frequency resolution with high deflection gain shall be not more than 200 MHz.

3.3.5.3 Frequency accuracy. Frequency accuracy with normal deflection gain shall be not more than 100 MHz.

3.3.5.3.1 Frequency accuracy with high deflection gain shall be not more than 200 MHz.

3.3.6 Characteristics applicable to all bands.

3.3.6.1 Dynamic range. Instantaneous dynamic range shall be not less than 60 dB with the RF gain control set for minimum attenuation.

3.3.6.2 Manual gain control. A manual gain control having not less than 50 dB of attenuation range shall be provided.

3.3.6.3 Minimum pulse-width. Minimum pulse width reception capability for full sensitivity shall be 150 nanoseconds (ns).

3.3.6.4 Maximum pulse-width. Maximum pulse width reception capability with not more than 5 percent sag (see 6.3) shall be not less than 50 microseconds (ms).

3.3.6.5 Video bandwidth. Video bandwidth shall be not less than 5 MHz.

3.3.6.6 Reject YIG filters. Reject YIG filters shall be provided in the 7-11 and 11-18 GHz ranges. (The 7-11 GHz filter shall be applicable to the 8.5-9.6 GHz band.) The YIG filters shall be tunable by the operator over the applicable band and beyond the band edges by not less than 50 MHz. A minimum of 40 dB rejection shall be provided. Bandwidth shall be 6 MHz minimum at the 30 dB rejection points and 45 MHz maximum at the 3 dB rejection points.

3.3.6.7 Bandwidth. The wide open bandwidth of the receivers shall be determined by fixed tuned filters preceeding the frequency discriminators. Filter characteristics are specified in Table III.

Table III. Fixed tuned filter characteristics.

Band	Frequency attenuation			
	≤1 dB		≥15 dB	≥35 dB
7-11 GHz	7 GHz to 11 GHz	+20 MHz -40 MHz +40 MHz -20 MHz	6.6 GHz 11.5 GHz	DC to 6 GHz 12 to 18 GHz

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Table III. Fixed tuned filter characteristics (continued).

Band	Frequency attenuation			
	≤1 dB		≥15 dB	≥35 dB
8.5-9.6 GHz	8.5 GHz to 9.6 GHz	+ 5 MHz -10 MHz +10 MHz - 5 MHz	8.4 GHz 9.7 GHz	DC to 8.2 GHz 9.9 to 18 GHz
11-18 GHz	11 GHz to 18 GHz	+30 MHz -60 MHz +60 MHz -30 MHz	10.4 GHz 18.6 GHz	DC to 9.5 GHz 19.5 to 26 GHz

3.3.6.8 Video interconnections. All video interfaces between units of the AN/WLR-11( ) and between the AN/WLR-11( ) and related equipments shall be via 50 ohm cables. Peaking adjustments shall be provided to compensate for cable high frequency loss.

3.3.6.9 Angle gating. An angle gating unit (AGU) shall be provided as an integral part of each control and display. This unit shall provide the operator with an effective frequency selection capability; since deflection angle is a function of frequency, signal gating by angles is equivalent to gating by frequency. It shall be capable of three modes of operation, OFF, ON and ALARM.

3.3.6.9.1 OFF mode. In the OFF mode it shall provide a single dashed radial cursor on the CRT. The POSITION control shall allow setting the cursor at any desired angular position. All signals on display over the entire band being received shall be available at the audio and video outputs.

3.3.6.9.2 ON mode. In the ON mode a second cursor shall be provided. The WIDTH control shall allow setting this cursor at any angle between 6 and 120 degrees clockwise with respect to the first cursor. Only these signals appearing on the display between the two cursors shall generate audio and external-going video.

3.3.6.9.3 ALARM mode. In the ALARM mode, signals appearing between the two cursors shall cause a Klaxon audio alarm lasting nominally 2.5 seconds.

3.4 Workmanship. Workmanship shall be in accordance with MIL-E-16400.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless

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ohms or less between the ... with the generator and the

disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will include: (a) Surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, (b) Government product inspection to measure quality of product to be offered for acceptance, and (c) Government product inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until performance of product to prescribed criteria has been demonstrated.

4.2 Classification of inspection. The method of examination and testing of the equipment shall be classified as follows:

(a) First article inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

(1) Production inspection (see 4.4.1)
(2) Burn-in test (see 4.4.2)
(3) Production control inspection (see 4.4.3)

(c) Reliability testing (see 4.5)
(d) Maintainability (see 4.6)
(e) Inspection for preparation for delivery (see 4.8)

4.3 First article inspection and test. First article inspection and test shall be made on one unit, unless otherwise specified (see 6.2). This inspection shall consist of examination and testing to determine compliance with the requirements of this specification and shall include the first article inspection requirements of MIL-E-16400 and the electrical performance tests. The following tests of MIL-E-16400 are not applicable:

(a) Water-cooling
(b) Inclination and noise
(c) Accelerated life
(d) Magnetic test

4.3.1 Burn-in test. The equipment submitted for first article inspection shall have passed 96 hours of continuous failure free burn-in at 40°C ambient temperatures with operational performance testing before, every 24 hours during, and after the burn-in test. The test documentation shall include each time a unit is submitted for burn-in, the failures which occur including a failure report and analysis for each failure and the performance parameters measured and

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their values. Any performance degraded below specification requirements shall be cause for stopping the tests, repairing the equipment, correcting the design as required, and restarting the test at zero time.

4.4 Quality conformance inspection.

4.4.1 Production inspection. The production inspection shall be in accordance with MIL-E-16400 and the equipment specification and shall be conducted on every equipment offered for delivery. The production inspection shall include a visual mechanical and an operational performance test as a minimum.

4.4.2 Burn-in test. Each equipment submitted for production inspection shall have passed 96 hours of continuous failure free burn-in at ambient room temperature with operational performance testing before, every 24 hours during, and after the burn-in test. The test documentation shall include each time a unit is submitted for burn-in, the failures which occur, the corrective actions taken, and the performance parameters measured and their values.

4.4.3 Production control inspection. This inspection shall be conducted on 1 out of the first 5, 1 out of the next 10 and 1 out of the next 15. This inspection shall be conducted on units that have passed production inspection. Tests shall be approved and consist of the following tests of 4.3 as a minimum:

(a) Weights and dimensions
(b) Supply line voltage and frequency variation only
(c) Heat test
(d) Operating tests

4.5 Reliability demonstration tests.

4.5.1 Reliability demonstration. The reliability demonstration tests shall be a fixed-time test of 3,876 equipment operating hours to demonstrate a 1,000 hour MTBF to a 90 percent lower confidence level. The environment for the test shall be in accordance with MIL-STD-781, Test Level A. Input voltage cycling will not be required. The reliability demonstration test shall be run 24 hours a day, seven days a week until the completion of 3,876 equipment operating hours. The power-on time shall be recorded by an elapsed time meter connected into the main power cable of the system. One hour per day may be used for minor preventative maintenance such as the adjustment of external controls or the replacement of panel lights. The on-off cycling shall consist of eleven and one-half hours on followed by a minimum of 1/2 hour off the cycle then being repeated for the duration of the test.

4.5.1.1 Failures. In the event of a failure, a failure report shall be generated indicating all the details of the failure, including the elapsed time meter reading at which the failure occurred, and a description of the repair action required to restore the system to proper operating condition. Upon satisfactory completion of the repair, the demonstration test shall be resumed. The test shall be run until the completion of 3,876 equipment operating hours regardless of the number of failures which occur. Upon completion of the test a final

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report shall be written which shall include copies of all failure reports plus other pertinent information, recommendations and conclusions.

4.5.1.2 Demonstrated reliability in terms of MTBF shall be determined by referring to Table IV.

Table IV. Demonstrated reliability in terms of MTBF.

Test time	No. of failures	Observed MTBF	MTBF at 90 percent Limited Confidence Level (LCL)
3,876	0	N/A	1,700
3,876	1	3,876	1,000
3,876	2	1,938	730
3,876	3	1,292	580
3,876	4	969	485
3,876	5	775	420
3,876	6	646	370
3,876	7	554	330
3,876	8	485	300

4.6 Maintainability demonstration test. A maintainability demonstration test shall be conducted on the AN/WLR-11( ) to evaluate its design limitations in terms of the median equipment repair time. The test shall be in accordance with MIL-STD-471, Method 4, test on the median (ERT) and the equipment shall demonstrate a median ERT of not greater than 0.5 hour. The techniques for meeting the 0.5 hour ERT shall be limited to replacement of the smallest replaceable (plug-in) assembly.

4.7 Electromagnetic compatibility test. The requirements of 3.1.13 shall be verified by tests performed in accordance with MIL-STD-461.

4.8 Inspection of preparation for delivery. Inspection of preparation for delivery shall be in accordance with Section 5.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing and marking. Unless otherwise specified, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing and marking specified in MIL-E-17555 (see 6.2).

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6. NOTES

6.1 Intended use. The unit described herein is intended for service on Navy ships of all types and classes, where it is expected to withstand continuous or intermittent use for long periods.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Number of first article samples to be submitted if other than as specified in 4.3.
(c) Preservation, packaging, packing, and marking required (see 5.1).

6.3 Definitions. For the purposes of this specification, the following definitions shall apply:

6.3.1 Minimum detectable (discernible) signal. That signal level in dBm at an RF input port which just causes intensification and minimum discernible spot deflection on the face of the cathode ray tube indicator of the control and display unit, with the original spot adjusted at the center and focused, then adjusted to a level just below visibility by the intensity control.

6.3.2 Tangential sensitivity. That signal level in dBm at an RF input port of a particular RF unit that causes the signal-plus-noise level as displayed on the face of a test oscilloscope to rise to a point where its baseline is exactly level with a line drawn at the peak of the original inherent unit noise.

6.3.3 Three-sixteenths inch (3/16 inch) (4.76 mm) deflection sensitivity. The signal level in dBm that will cause a three-sixteenths of an inch (3/16 inch) (4.76 mm) deflection of the spot shown by the intensification of the CRT beam caused by the application of a minimum discernible signal.

6.3.4 Burn-in. Operation at ambient room temperature for the specified number of hours.

6.3.5 Wide-open channel. The wide-open channel (w.o.) is defined as the full band width or pass-band of one of the three bands; 7 to 11 GHz, 8.5 to 9.6 GHz or 11 to 18 GHz.

6.3.6 Narrow-band channel. The narrow-band or tuned radio frequency (TRF) channel is defined as that signal path containing the bandpass or accept YIG. This channel, of limited bandwidth, may be tuned over the full width of the wide-open channel.

6.3.7 Normal deflection gain. The deflection obtained with the operator-controlled deflection gain switch in the normal position. Display deflection sensitivity of both the X and Y video circuits are then to be 2.2 ±.1 inches per volt.

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6.3.8 High deflection gain. The deflection obtained with the operator controlled deflection gain switch in the high position. Display deflection sensitivity of both the X and Y video circuits are then to be 8.8 ±.4 inches per volt.

6.3.9 Pulse sag. Pulse sag or droop is defined as the amount, in percent, that the top of a pulse falls below that of its leading edge during the pulse period.

6.4 First article. Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.5 Supersession data. This specification includes the requirements of Naval Electronic Systems Command Contract Specification ELEX-R-167 (Receiving Set, Electronic Countermeasures AN/WLR-11( )) dated 7 December 1973.

Preparing activity:
Navy - EC

(Project 5895-N242)

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